

EP BrainHealth 2026 — Call 2: Neurodegeneration

Construction of a Multimodal Causal Knowledge Graph Powered by Medical Imaging and Brain Signals for the Monitoring of Neurodegenerative Diseases

Project Acronym: **CauAD**

Partner	Country	Principal Investigator	Primary Role
Galatasaray University (Coordinator)	Türkiye	Dr. Sultan N. Turhan	KG engineering, ontology co-design, causal discovery, coordination
CY Cergy Paris Université (ETIS Lab)	France	Dr. Redouane Bouhamoum	Ontology design, semantic integration, AlzKB validation
Università Politecnica delle Marche (Br3in Lab)	Italy	Prof. Laura Burattini	Neuroimaging, brain signals, metabolic modeling, biomarker validation

Call: EP BrainHealth JTC2026 Call 2 — Neurodegeneration | **Duration:** 36 months

Pre-Proposal Deadline: 10 March 2026 (14:00 CET) | **Full Proposal:** 30 June 2026

Countries: 3 (Türkiye, France, Italy) | **Türkiye underrepresented:** Yes (allows up to 7 partners)

National Funding Agencies: TÜBİTAK (TR), ANR (FR), MUR (IT)

This document presents a comprehensive project outline for the CauAD consortium, detailing partner roles, specific task assignments, deliverables, and the publication plan. It is intended to support internal alignment among all partners prior to pre-proposal submission.

Table of Contents

- 1. Project Rationale and UNIVPM Contribution
- 2. Research Questions
- 3. Alignment with Call Requirements
- 4. Consortium and Expertise
- 5. Five-Phase Work Plan with Partner Assignments
- 6. UNIVPM: Tasks, Deliverables, and Team Mapping
- 7. Workload Distribution and Effort Estimates
- 8. Timeline and Milestones
- 9. Expected Outputs and Publication Plan
- 10. Next Steps and Action Items

1. Project Rationale and UNIVPM Contribution

The CauAD project constructs a semantically-grounded Knowledge Graph from the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset and applies causal discovery algorithms to characterise the biomarker cascade underlying Alzheimer's disease progression. The project builds upon an operational Neo4j Knowledge Graph (~407,000 nodes, ~1.16 million relationships) previously validated and published at IEEE Big Data 2025, which demonstrated 100% pixel-value preservation for all MRI and PET modalities and millisecond-level metadata retrieval via Elasticsearch.

The EP BrainHealth 2026 Call 2 requires investigation of neurodegeneration across **at least two factor categories** (biological, lifestyle/social, or environmental) and at least two of three **research aspects** (mechanisms, diagnosis/prognosis, prevention/intervention). To meet these requirements and substantially extend our existing infrastructure, we propose integrating **functional and structural brain connectivity measures** (fMRI, DTI) and **metabolic comorbidity status** into the Knowledge Graph as new causal variables, capabilities that require domain expertise in neuroimaging signal processing and metabolic systems modeling.

1.1 Why Br3in Lab Is the Right Partner

Our consortium possesses strong capabilities in knowledge graph engineering (Galatasaray and CY Cergy Paris, jointly), but has a critical gap in three areas:

- **Neuroimaging signal processing** : fMRI functional connectivity analysis, DTI tractography, PET quantification, and MRI volumetric extraction require specialised expertise in brain signal and image processing. Researchers and students bring precisely this expertise.
- **Metabolic systems modeling** : Investigating the causal role of Type 2 diabetes and insulin resistance in neurodegeneration requires mathematical modeling of glucose-insulin dynamics. Dr. Morettini's background in the EU FP7 MISSION-T2D project provides this capability, which neither Galatasaray nor CY Cergy possesses.
- **Biostatistical validation** : Longitudinal trajectory modeling and statistical quality control of extracted biomarker features require rigorous biostatistical methodology. Dr. Sbröllini's expertise in AI-assisted clinical decision support addresses this need.

In summary: UNIVPM leads all neuroimaging processing, metabolic variable analysis, longitudinal trajectory modeling, and biological validation of causal results. UNIVPM is not responsible for Knowledge Graph engineering, causal algorithm implementation, or ontology design; those are jointly managed by Galatasaray and CY Cergy.

2. Research Questions

Primary Research Question: Can the causal structure of Alzheimer's disease biomarker progression, ranging from amyloid accumulation and tau propagation to neurodegeneration and cognitive decline, be effectively recovered and quantitatively characterized by applying constraint-based and score-based causal discovery algorithms to multimodal features extracted from an ontology-grounded Knowledge Graph that integrates clinical, CSF biomarker, neuroimaging, and genetic data from the ADNI cohort?

Secondary Research Question: Does the inclusion of metabolic comorbidity status (Type 2 diabetes, insulin resistance markers, vascular risk factors) as a structured causal variable in the Knowledge Graph significantly modify the inferred disease trajectory, revealing mechanistic pathways between metabolic dysfunction and accelerated neurodegeneration?

This formulation maps directly to the call's requirements: the primary research question addresses **biological factors** (genetics, CSF biomarkers, brain circuits and connectivity), while the secondary question addresses **lifestyle and social factors** (metabolic comorbidity as a modifiable risk factor linked to diet, physical activity, and cardiometabolic health). Together, they span all three research aspects: **mechanisms of action** (causal pathway discovery), **early recognition and diagnosis** (connectivity biomarkers for preclinical detection), and **prevention and intervention** (metabolic risk factors as modifiable targets).

3. Alignment with Call Requirements

3.1 Factor Categories

Factor Category	How CauAD Addresses It	ADNI Data Sources
Biological Factors (genetics, -omics, circuits, vascular)	APOE genotype; CSF biomarkers (A β 42, tau, pTau); amyloid/tau PET SUVR; MRI volumetrics; fMRI functional connectivity; DTI white matter integrity	APOERES, UPENNBIOMK, UCSFFSX7, UCBERKELEY_*, fMRI DICOM, DTIROI
Lifestyle and Social Factors (metabolic comorbidity, education, sex/gender)	T2D status, diabetes medications, fasting glucose, metabolic syndrome estimation; education and sex as confounders in causal models	LABDATA, MEDHIST, BACKMEDS, PTDEMOG
Environmental Factors (co-morbidities, multi-site protocol effects)	Medical history, adverse events; multi-site protocol variation modeled as latent confounder (FCI algorithm handles this)	MEDHIST, ADVERSE, ROSTER (site and phase)

3.2 Research Aspects

Research Aspect	Project Coverage	Lead Partner
1. Mechanisms of action	Causal discovery identifies directed mechanistic pathways (A→T→N); fMRI/DTI reveal circuit-level mechanisms; metabolic modifier analysis reveals T2D → neurodegeneration pathways	GSU (algorithms) UNIVPM (imaging and metabolism)
2. Early recognition, diagnosis, prognosis	ATN biological staging via PET/CSF; brain connectivity biomarkers for preclinical detection; causal graph enables prognosis via upstream biomarker status	UNIVPM (biomarkers) GSU (KG queries)
3. Prevention, treatments, interventions	Metabolic risk factors identified as modifiable causal targets; AlzKB drug-target cross-validation; DoWhy causal effect estimation of potential interventions	UNIVPM (metabolic) CYT (AlzKB drugs)

4. Consortium and Expertise

Partner 1: Galatasaray University (Coordinator) — Türkiye

PI: Dr. Sultan N. Turhan | **Co-PI:** Asst. Prof. Özgün Pınarer | **Researcher:** Oğuzhan Güngör (MSc candidate, lead developer)

Infrastructure: Operational Neo4j KG (~407K nodes, ~1.16M relationships), Elasticsearch, Redis. Published at IEEE Big Data 2025. 16-step ETL pipeline covering 2,638 patients across 30,267 visits from ADNI1/2/GO/3/4.

Role: Project coordination; Knowledge Graph engineering and maintenance; Ontology extension (metabolic and lifestyle concept modules); causal algorithm implementation (PC, FCI, GES, DAG-GNN); DoWhy causal inference; feature matrix extraction and integration; financial and technical reporting; AlzKB cross-validation of discovered causal edges; ICD-10/SNOMED-CT disease classification; web-based query interface for the Causal KG; lead authorship on KG architecture methodology paper.

Partner 2: CY Cergy Paris Université (ETIS Lab) — France

PI: Dr. Redouane Bouhamoum | **Co-PIs:** Dr. Souhila Arib, Dr. Hajer Baazaoui

Expertise: Ontology engineering, semantic web technologies, disease classification systems, SPARQL reasoning.

Role: Ontology extension (metabolic and lifestyle concept modules); background knowledge encoding for causal algorithms; AlzKB cross-validation of discovered causal edges; ICD-10/SNOMED-CT disease classification; web-based query interface for the Causal KG; lead authorship on KG architecture methodology paper.

Partner 3: Università Politecnica delle Marche (Br3in Lab) — Italy

PI: Prof. Laura Burattini (PhD, University of Rochester, USA)

Key Team Members:

- Dr. Micaela Morettini, glucose-insulin regulatory modeling, Type 2 Diabetes (EU FP7 MISSION-T2D)
- Researcher 2, brain signal and image processing, cortical connectivity (EEG, fMRI, DTI)
- Researcher 3 Biostatistics, AI for clinical decision support, signal analysis
- Student 1, neuroimaging algorithms, brain connectivity characterisation
- Student 2, MRI for neurodegenerative diseases, pathological pattern detection
- Student 3, fMRI/DTI processing pipelines, structural connectivity

Role: All neuroimaging processing (resting-state fMRI, DTI tractography, amyloid/tau PET quantification, MRI volumetrics); metabolic comorbidity analysis; longitudinal trajectory modeling; biological validation and interpretation of discovered causal pathways.

5. Five-Phase Work Plan with Partner Assignments

GSU = Galatasaray University | CYT = CY Cergy Paris (ETIS Lab) | UNIVPM = Br3in Lab | J = Joint

Phase 1 — Foundation (Months 1-6)

Task	Description	Lead	Support
T1.1	Consortium setup, data sharing agreements, ADNI access confirmation for all partners	GSU	J
T1.2	AD-DPC ontology extension: metabolic and physical activity concept modules	CYT	GSU
T1.3	Metabolic variables extraction from ADNI (T2D status from MEDHIST; fasting glucose from LABDATA; diabetes medications from BACKMEDS; BMI/BP from VITALS) and mapping to ontology	GSU	CYT
T1.4	KG v2: integration of metabolic and lifestyle nodes into existing Neo4j infrastructure	GSU	CYT
T1.5	fMRI, DTI, and PET data inventory and quality assessment across ADNI cohort; feasibility report on imaging subcohort size	UNIVPM	GSU

Phase 2 — Data Integration (Months 6-14)

Task	Description	Lead	Support
T2.1–2.5	Resting-state fMRI preprocessing pipeline (motion correction, bandpass filtering, nuisance regression)	UNIVPM	—
T2.2	Functional connectivity matrix extraction per patient per visit (default mode network, frontoparietal, limbic networks)	UNIVPM	—
T2.3	DTI tractography and white matter integrity metrics (FA, MD) for key tracts (cingulum, fornix, uncinate fasciculus)	UNIVPM	—
T2.4	Amyloid and Tau PET quantification: SUVR computation, Centiloid conversion, regional uptake patterns	UNIVPM	GSU
T2.5	Longitudinal trajectory modeling: per-patient connectivity slope computation via linear mixed-effects models	UNIVPM	GSU
T2.6	Registration of neuroimaging and metabolic temporal features into the Knowledge Graph	GSU	UNIVPM
T2.7	Unified feature matrix extraction from integrated KG (biomarker + genetic + metabolic + connectivity variables)	GSU	J

Note: Resting-state fMRI was introduced in ADNI-2/GO and DTI coverage varies across phases. T1.5 will produce a precise inventory of available imaging subjects. The causal analysis will use the full cohort (~2,600 subjects) for CSF, cognitive, and genetic variables, and a nested imaging subcohort (estimated 500-800 subjects with fMRI; 400-600 with DTI) for connectivity features. This nested design follows established practice in ADNI multimodal studies.

Phase 3 — Causal Discovery (Months 14-24)

Task	Description	Lead	Support
T3.1	Background knowledge encoding: ontological temporal ordering constraints for causal algorithms (e.g. amyloid before tau)	CYT	GSU
T3.2	PC algorithm execution on integrated feature matrix	GSU	—
T3.3	FCI algorithm execution (robust to latent confounders from unobserved variables)	GSU	—
T3.4	GES/FGES algorithm execution (score-based comparison approach)	GSU	—
T3.5	Cross-algorithm reconciliation: high-confidence edge identification (≥ 2 algorithm agreement)	GSU	J
T3.6	Temporal Causal DAG v1 construction	GSU	J

Phase 4 — Validation and Inference (Months 24-30)

Task	Description	Lead	Support
T4.1	A→T→N cascade recovery test: benchmark discovered graph against established biomarker ordering (Jack et al. 2010)	GSU	UNIVPM
T4.2	DoWhy causal effect estimation for key edges (e.g. amyloid positivity → MMSE decline rate)	GSU	—
T4.3	Refutation tests: placebo treatment, random common cause, data subset permutations	GSU	—
T4.4	Metabolic modifier analysis: causal contribution of T2D and insulin resistance to the neurodegeneration pathway	UNIVPM	GSU
T4.5	AlzKB cross-validation: map discovered causal edges to AlzKB entity pairs (118,902 entities, 1.3M relationships)	CYT	GSU
T4.6	Expert neurologist review workshop: structured evaluation of Causal DAG v1	GSU	J
T4.7	Graph refinement based on expert feedback → Temporal Causal DAG v2 (validated)	J	—
T4.8	CAUSES relationship embedding into Neo4j with confidence scores, algorithm provenance, and literature annotations	GSU	J

Phase 5 — Dissemination and Closure (Months 30-36)

Task	Description	Lead	Support
T5.1	Manuscript 1: KG architecture and ontology extension methodology	CYT	GSU
T5.2	Manuscript 2: Neuroimaging pipeline and biomarker extraction methodology	UNIVPM	GSU

Task	Description	Lead	Support
T5.3	Manuscript 3: Causal discovery findings and temporal causal graph results	GSU	UNIVPM
T5.4	Open-source release: KG dump, causal pipeline, fMRI/DTI processing scripts (GitHub)	GSU	UNIVPM
T5.5	Web query interface for the Causal KG	CYT	GSU
T5.6	Conference presentations at AAIC or equivalent AD-focused venue	J	—
T5.7	Final technical and financial reporting to TÜBİTAK, ANR, MUR	GSU	J
T5.8	Midterm symposium presentation (~Month 20; EP BrainHealth consortium obligation)	GSU	J

6. UNIVPM: Tasks, Deliverables, and Team Mapping

UNIVPM leads 8 tasks and contributes to 5 additional tasks across all project phases. Estimated total effort: **approximately 40 person-months** over 36 months.

6.1 UNIVPM-Led Tasks and Deliverables

Task	Description	P h.	P M	Deliverable
T1.5	fMRI/DTI/PET data inventory, quality assessment, feasibility report	1	3	D1.5: Data inventory report
T2.1	Resting-state fMRI preprocessing pipeline (motion, filtering, regression)	2	21	D2.1–2.5: Neuroimaging feature package
T4.4	Metabolic modifier analysis: causal contribution of T2D/insulin resistance to neurodegeneration	4	5	D4.4: Metabolic modifier report
T5.2	Lead manuscript: neuroimaging pipeline and biomarker extraction methodology	5	4	Journal paper (UNIVPM lead author)

6.2 UNIVPM Support Contributions

- T2.6 — Provide neuroimaging feature specifications for KG registration (GSU integrates)
- T4.1 — Biological validation of recovered A→T→N cascade against neuroimaging evidence
- T5.3 — Co-author causal discovery findings manuscript
- T5.4 — Contribute fMRI/DTI processing scripts to open-source release
- T5.6 — Conference co-presentations

6.3 Team-to-Task Mapping

Team Member	Expertise	Primary Tasks	Est. Effort
Prof. Laura Burattini (PI)	Biomedical data processing; PI oversight; industry experience	Overall UNIVPM coordination; T4.4 oversight; T5.2 senior author; expert workshops	~6 PM
Researcher 1	Brain signal/image processing; cortical connectivity (EEG, fMRI, DTI)	T2.1 (fMRI pipeline lead); T2.2 (FC extraction); T2.3 (DTI tractography)	~8 PM
Dr. Micaela Morettini	Glucose-insulin regulatory modeling; Type 2 Diabetes; EU FP7 MISSION-T2D	T4.4 (metabolic modifier analysis, core expertise); T1.5 (metabolic data review)	~5 PM
Researcher 2	Biostatistics; AI for signal analysis; clinical decision support	T2.5 (mixed-effects models); statistical QC across all feature extraction tasks	~5 PM
Student 1	Neuroimaging algorithms; EEG/MRI processing; brain connectivity	T2.1/T2.2 (hands-on fMRI processing); T2.4 (PET)	~6 PM
Student 2	MRI for neurodegenerative diseases; AI for pathological pattern detection	T1.5 (data inventory); T2.4 (PET quantification)	~5 PM
Student 3	fMRI/DTI pipeline development; structural and functional connectivity	T2.3 (DTI tractography); T2.1 (fMRI co-development)	~5 PM

7. Workload Distribution and Effort Estimates

Partner	Lead Tasks	Support Tasks	Estimated Effort	Primary Responsibility
GSU (Coordinator)	19 tasks (T1.1, T1.3, T1.4, T2.6, T2.7, T3.2–3.6, T4.1–4.3, T4.6, T4.8, T5.3, T5.4, T5.7, T5.8)	10 tasks	~54 PM	KG engineering, ontology causal algorithms, coordination
CYT (France)	5 tasks (T1.2, T3.1, T4.5, T5.1, T5.5)	8 tasks	~30 PM	Ontology design, AlzKB validation, web interface
UNIVPM (Italy)	8 tasks (T1.5, T2.1–T2.5, T4.4, T5.2)	5 tasks	~40 PM	Neuroimaging, metabolic modeling, biological validation
Joint	2 tasks (T4.7, T5.6)	—	Included above	Consensus activities

The distribution reflects each group's core expertise: Galatasaray handles data engineering, Knowledge Graph infrastructure, and causal algorithm implementation; CY Cergy and Galatasaray manage the ontology and semantic layer, external validation against AlzKB, and the web query interface; UNIVPM provides all neuroimaging processing, metabolic domain expertise, and biological validation. Each partner leads a distinct and substantive contribution to the project.

8. Timeline and Milestones

Phase	Months	Key Milestone	UNIVPM Activity
1: Foundation	M1–6	KG v2 with metabolic nodes; ontology extension complete; ADNI access for all partners	Data inventory (T1.5); assess fMRI/DTI/PET coverage and feasibility
2: Integration	M6–14	All neuroimaging features extracted and registered in the Knowledge Graph	fMRI pipeline (T2.1–2.2); DTI processing (T2.3); PET quantification (T2.4); trajectory modeling (T2.5)
3: Discovery	M14–24	Temporal Causal DAG v1 constructed; cross-algorithm consensus achieved	Contribute to biological interpretation of edges; provide feature QC reports
4: Validation	M24–30	Validated DAG v2; DoWhy effects estimated; AlzKB cross-referenced	Metabolic modifier analysis (T4.4); expert workshop; biological validation of causal pathways
5: Dissemination	M30–36	3+ publications submitted; open-source release; final reports filed	Lead neuroimaging paper (T5.2); co-author causal paper; conference presentations

9. Expected Outputs and Publication Plan

9.1 Deliverables

ID	Deliverable	Lead	Due
D1.5	Neuroimaging data inventory and feasibility report	UNIVPM	M6
D2.1–2.5	fMRI preprocessing pipeline and QC report	UNIVPM	M6–M14
D2.7	Unified integrated feature matrix from KG	GSU	M14
D3.6	Temporal Causal DAG v1	GSU	M24
D4.4	Metabolic modifier analysis report	UNIVPM	M28
D4.5	AlzKB cross-validation report	CYT	M30
D4.7	Validated Temporal Causal DAG v2 (final)	Joint	M30
D5.1–5.3	3+ journal publications	All	M36
D5.4	Open-source code release (GitHub)	GSU	M36
D5.5	Web query interface for Causal KG	Joint	M36

9.2 Publication Plan

#	Working Title	Target Venue	Lead	Timeline
P1	Ontology-driven KG architecture for multimodal AD data integration	Journal of Biomedical Informatics	CYT	M18–24
P2	Neuroimaging biomarker extraction pipeline for causal models of AD	NeuroImage or Alzheimer's & Dementia	UNIVPM	M18–24
P3	Causal discovery in the AD biomarker cascade: from Knowledge Graph to mechanistic insight	Brain Communications or npj Digital Medicine	GSU	M24–30
P4	Metabolic comorbidity as a causal modifier of AD neurodegeneration: evidence from ADNI	Frontiers in Aging Neuroscience	UNIVPM + GSU	M28–34

UNIVPM holds lead authorship on P2 and co-lead authorship on P4, providing high-impact publication opportunities in neuroimaging and translational neuroscience venues.

10. Next Steps and Action Items

Step	Action	Responsible	Deadline
1	UNIVPM reviews this document and confirms interest; flags any tasks to adjust or add	UNIVPM	1 March
2	Joint video call (all three partners): finalise work packages, refine RQs, divide writing	GSU organises	3–4 March
3	Each partner drafts their pre-proposal section. UNIVPM: Phase 2 description + team CVs	All partners	7 March
4	Pre-proposal submission to EP BrainHealth Call 2 secretariat	GSU	10 March (14:00 CET)
5	TÜBİTAK national submission (Turkish deadline)	GSU	16 March
6	If invited: co-write full proposal with detailed methodology, budget, and Gantt chart	All partners	30 June

Key Dates:
Pre-Proposal Deadline: **10 March 2026** (14:00 CET)
TÜBİTAK National Submission: **16 March 2026**
Full Proposal (if invited): **30 June 2026**
Expected Project Start: **Early-Mid 2027** | Duration: **36 months**

Budget guidance: EP BrainHealth funds through synchronised national grants — each partner is funded by their own country’s agency (MUR for Italy, ANR for France, TÜBİTAK for Türkiye). Each partner typically requests €100,000–200,000 for a 3-year project, covering personnel, travel, computing infrastructure, and consumables. Specific Italian funding rules and eligible costs under MUR should be discussed on the joint call.

We are confident that the Br3in Lab’s expertise in brain signal processing, neuroimaging analysis, and metabolic systems modeling makes UNIVPM the ideal partner for producing biologically credible causal models of Alzheimer’s disease progression. We look forward to your feedback and to building this proposal together.

With kind regards,

Dr. Sultan Nezihe Turhan , Principal Investigator, Galatasaray University
Asst. Prof. Özgün Pınarer, Co-Investigator, Galatasaray University
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